Atmospheric entry

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use energy and geometry to discuss entry heating without claiming a guaranteed outcome.

SUCCESS CRITERIA - I can...

1. Compare kinetic energy with a stated heating requirement.
2. Explain the role of surface area relative to mass.
3. Name missing factors in a survival prediction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “ablation” and “thermal energy” to explain your model or result. Sentence starters are optional.

ablation: loss of surface material during heating or erosion

thermal energy: internal energy associated with microscopic motion/interactions

impact: a collision or strike between objects

01 Bring back an earlier idea

A process has $\Delta U = -15$ J and $W_{by} = 21$ J.
Find Q and decide whether heat enters or leaves.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $K = \frac{1}{2}mv^2$
$Q_{absorbed}$ = fraction absorbed × energy available.
Entry also involves ablation and fragmentation. An energy comparison alone cannot predict survival.

Worked reasoning: A fictional entering object loses 1000 J of mechanical energy; a model assigns 700 J to heated air and 300 J to the object. Those shares are supplied assumptions, not universal atmospheric-entry fractions.

Your turn: Draw the energy budget and identify one measurement needed to test the shares.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Compare a small and a large sphere of the same material. Which has more surface area per unit mass? What does that suggest about surface interactions per kilogram?

Sentence starter: Both ... . They differ in ... because ... .



A B

Marzano check - circle: ① ② ③

04 Solve and justify

For a hypothetical 10 kg rock at 15 km/s, compute K. A supplied heating/vaporization requirement is 8.6×10^6 J/kg.
Find total requirement and energy ratio.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Name at least three additional quantities needed before claiming a particular rock completely ablates or reaches the ground.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Choose a method

For fixed GM, launch radius becomes 16 times as large. **Compare** ideal escape speeds and name the equation.

Word help: radius = distance from the center to the edge of a circle or sphere

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

07 Exit check

If radius doubles at fixed density, derive the area-to-mass multiplier and explain why kinetic-energy-per-mass alone cannot predict a size threshold for survival.

Word help: derive = show the reasoning or algebra that produces a relationship

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 2 kg sample has 20 kJ deposited as internal energy. With effective c=500 J/(kg·K), estimate ΔT if no phase change or loss occurs.
State why this is not a survival threshold.

Word help: estimate = find an approximate value

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3